

Investigación
de Operaciones

Branch & Bound

Solving integer programming problems

Dayron Josué Porras Pérez (2023124693)
Javier Lee Liang (2020184824)
José Julián Araya Castillo (2020098909)

Integer Programming

November 8, 2026

Contents

Methodology

Branch & Bound algorithm

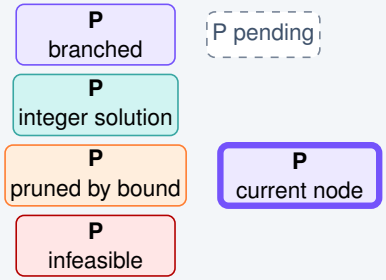
For each subproblem

1. Solve its LP relaxation with **LINGO**.
2. **Fathom** the node if the relaxation is infeasible, if its solution is integer (update the incumbent z^*), or if it does not improve z^*.
3. Si no se poda, **branch** on the first fractional variable $x_j = v$:
   $x_j \leq \lfloor v \rfloor$ y $x_j \geq \lceil v \rceil$.

Conventions

Depth-first search (the $\leq$ branch is explored first).
Subproblems are numbered in order of creation and the *step* gives the order of resolution.

Tree legend



Problem 1

Problem 1 · Statement

$$\begin{aligned} &\max\ z = 5x_1 + 2x_2 \\ \text{s.t.}\ \ &3x_1 + x_2 \le 12 \\ &x_1 + x_2 \le 5 \\ &x_1, x_2 \ge 0;\ \ x_1, x_2 \text{ integer} \end{aligned}$$

Problem 1 · LINGO: root LP (P1)

LINGO model

```
MAX = 5*x1 + 2*x2;
3*x1 + x2 <= 12;
x1 + x2 <= 5;
END
```

Result

$z = 20.5, \quad x_1 = 3.5, \quad x_2 = 1.5$

Decision

$x_1 = 3.5$ is fractional $\Rightarrow$ branch: **P2** with $x_1 \leq 3$ and **P3** with $x_1 \geq 4$.

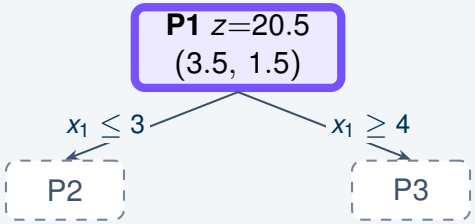
LINGO screenshot

save as `lingo/p1_P1.png`

Problem 1 · Tree after solving P1

Step 1: P1 solved (branched).

Current incumbent: none



Problem 1 · Subproblem P2: LINGO

LINGO model

```
MAX = 5*x1 + 2*x2;
3*x1 + x2 ≤ 12;
x1 + x2 ≤ 5;
x1 ≤ 3;
END
```

Result

$z = 19, \quad x_1 = 3, \quad x_2 = 2$

Decision

Feasible integer solution $\Rightarrow$ new **incumbent** $z^* = 19$. Node fathomed.

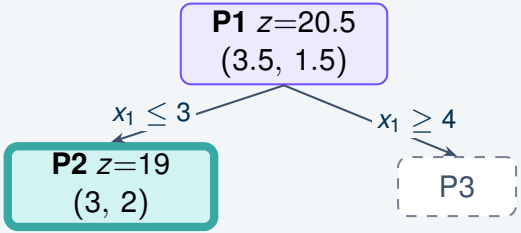
LINGO screenshot

save as `lingo/p1_P2.png`

Problem 1 · Tree after solving P2

Step 2: P2 solved (integer).

Current incumbent: $z^* = 19$ at $(3, 2)$



Problem 1 · Subproblem P3: LINGO

LINGO model

```
MAX = 5*x1 + 2*x2;
3*x1 + x2 <= 12;
x1 + x2 <= 5;
x1 >= 4;
END
```

Result

$z = 20, \quad x_1 = 4, \quad x_2 = 0$

Decision

Feasible integer solution $\Rightarrow$ new **incumbent** $z^* = 20$. Node fathomed.

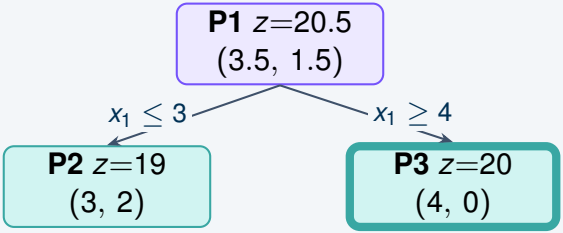
LINGO screenshot

save as `lingo/p1_P3.png`

Problem 1 · Tree after solving P3

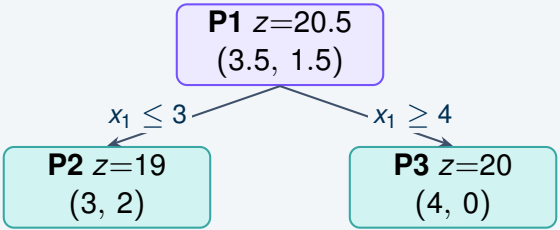
Step 3: P3 solved (integer).

Current incumbent: $z^* = 20$ at $(4, 0)$



Problem 1 · Final solution

Optimal solution: $z^* = 20$, $x_1^* = 4$, $x_2^* = 0$ (3 subproblems solved)



Problem 3

Problem 3 · Statement

$$\begin{aligned} \max\ & z = 2x_1 + 3x_2 \\ \text{s.t.}\ & x_1 + 2x_2 \le 10 \\ & 3x_1 + 4x_2 \le 25 \\ & x_1, x_2 \ge 0;\ \ x_1, x_2 \text{ integer} \end{aligned}$$

Problem 3 · LINGO: root LP (P1)

LINGO model

```
MAX = 2*x1 + 3*x2;
x1 + 2*x2 ≤ 10;
3*x1 + 4*x2 ≤ 25;
END
```

Result

$z = 17.5, \quad x_1 = 5, \quad x_2 = 2.5$

Decision

$x_2 = 2.5$ is fractional $\Rightarrow$ branch: **P2** with $x_2 \leq 2$ and **P3** with $x_2 \geq 3$.

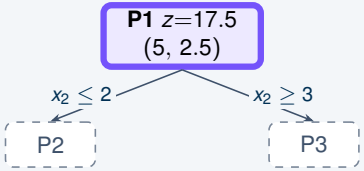
LINGO screenshot

save as `lingo/p3_P1.png`

Problem 3 · Tree after solving P1

Step 1: P1 solved (branched).

Current incumbent: none



Problem 3 · Subproblem P2: LINGO

LINGO model

```
MAX = 2*x1 + 3*x2;
x1 + 2*x2 ≤ 10;
3*x1 + 4*x2 ≤ 25;
x2 ≤ 2;
END
```

Result

$z = \frac{52}{3} \approx 17.33, \quad x_1 = \frac{17}{3} \approx 5.66, \quad x_2 = 2$

Decision

$x_1 = \frac{17}{3} \approx 5.66$ is fractional $\Rightarrow$ branch: **P4** with $x_1 \leq 5$ and **P5** with $x_1 \geq 6$.

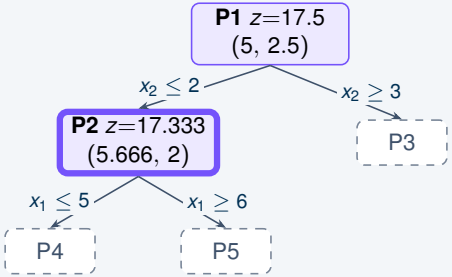
LINGO screenshot

save as `lingo/p3_P2.png`

Problem 3 · Tree after solving P2

Step 2: P2 solved (branched).

Current incumbent: none



Problem 3 · Subproblem P4: LINGO

LINGO model

```
MAX = 2*x1 + 3*x2;
x1 + 2*x2 ≤ 10;
3*x1 + 4*x2 ≤ 25;
x1 ≤ 5;
END
```

Result

$z = \frac{35}{2} \approx 17.5, \quad x_1 = 5, \quad x_2 = \frac{5}{2} \approx 2.5$

Decision

Subproblem **P4** is the same as **P1**, so it gets pruned, because it is a repeated subproblem.

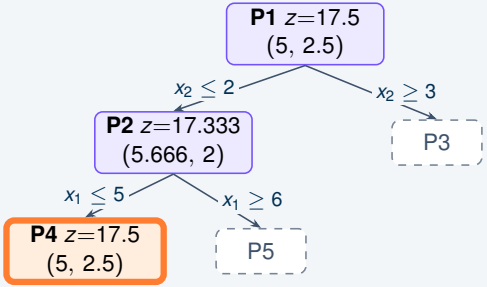
LINGO screenshot

save as `lingo/p3_P4.png`

Problem 3 · Tree after solving P4

Step 3: P4 solved (pruned).

Current incumbent: none



Problem 3 · Subproblem P5: LINGO

LINGO model

```
MAX = 2*x1 + 3*x2;
x1 + 2*x2 ≤ 10;
3*x1 + 4*x2 ≤ 25;
x1 ≥ 6;
END
```

Result

$z = \frac{69}{4} \approx 17.25, \quad x_1 = 6, \quad x_2 = \frac{7}{4} \approx 1.75$

Decision

$x_2 = \frac{7}{4} \approx 1.75$ is fractional $\Rightarrow$ branch: **P6** with $x_2 \leq 1$ and **P7** with $x_2 \geq 2$.

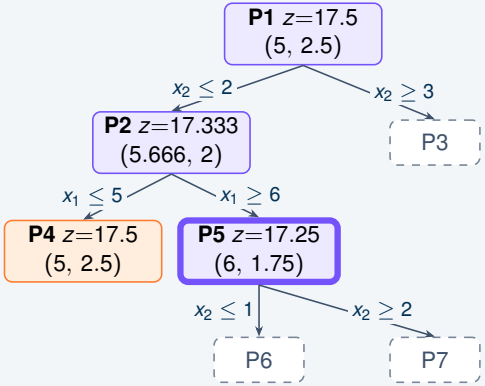
LINGO screenshot

save as `lingo/p3_P5.png`

Problem 3 · Tree after solving P5

Step 4: P5 solved (branched).

Current incumbent: none



Problem 3 · Subproblem P6: LINGO

LINGO model

```
MAX = 2*x1 + 3*x2;
x1 + 2*x2 ≤ 10;
3*x1 + 4*x2 ≤ 25;
x2 ≤ 1;
END
```

Result

$z = 17, \quad x_1 = 7, \quad x_2 = 1$

Decision

Feasible integer solution $\Rightarrow$ new **incumbent** $z^* = 17$. Node fathomed.

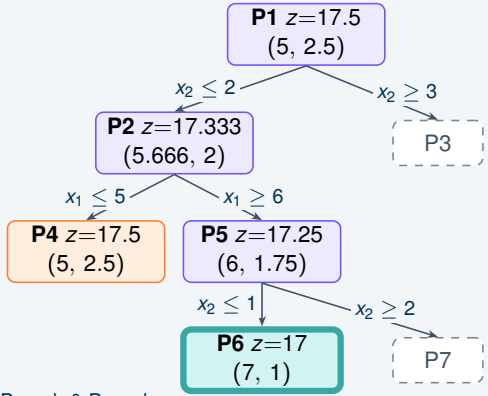
LINGO screenshot

save as `lingo/p3_P6.png`

Problem 3 · Tree after solving P6

Step 5: P6 solved (integer).

Current incumbent: $z^* = 17$ at $(7, 1)$



Problem 3 · Subproblem P7: LINGO

LINGO model

```
MAX = 2*x1 + 3*x2;
x1 + 2*x2 ≤ 10;
3*x1 + 4*x2 ≤ 25;
x2 ≥ 2;
END
```

Result

$z = \frac{35}{2} \approx 17.5, \quad x_1 = 5, \quad x_2 = \frac{5}{2} \approx 2.5$

Decision

Subproblem **P7** is the same as **P1**, so it gets pruned, because it is a repeated subproblem.

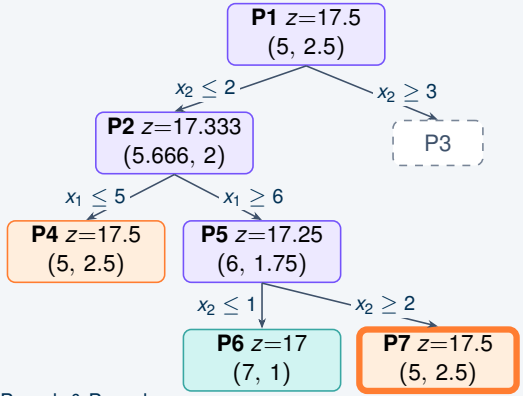
LINGO screenshot

save as `lingo/p3_P7.png`

Problem 3 · Tree after solving P7

Step 6: P7 solved (pruned).

Current incumbent: $z^* = 17$ at $(7, 1)$



Problem 3 · Subproblem P3: LINGO

LINGO model

```
MAX = 2*x1 + 3*x2;
x1 + 2*x2 ≤ 10;
3*x1 + 4*x2 ≤ 25;
x2 ≥ 3;
END
```

Result

$z = 17, \quad x_1 = 4, \quad x_2 = 3$

Decision

Feasible integer solution, but the current incumbent solution has the same Z value.

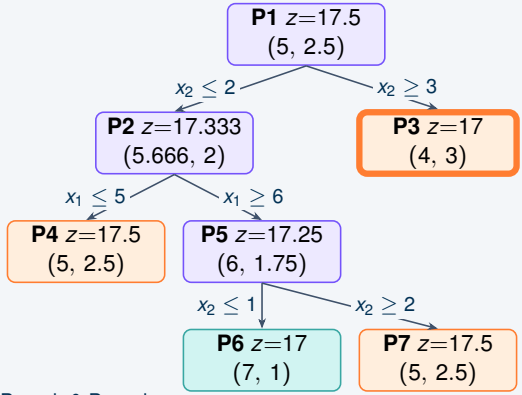
LINGO screenshot

save as `lingo/p3_P7.png`

Problem 3 · Tree after solving P3

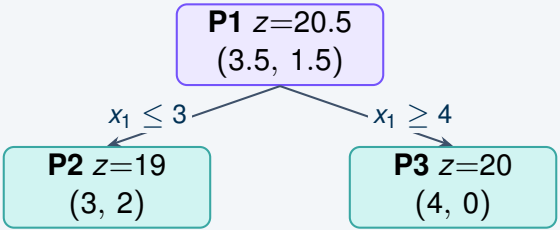
Step 6: P3 solved (pruned).

Current incumbent: $z^* = 17$ at $(7, 1)$



Problem 1 · Final solution

Optimal solution: $z^* = 17$, $x_1^* = 7$, $x_2^* = 1$ (7 subproblems solved)



Problem 4

Problem 4 · Statement

$$\begin{aligned} &\max\ z = 4x_1 + 3x_2 \\ \text{s.t.}\ \ &4x_1 + 9x_2 \le 26 \\ &8x_1 + 5x_2 \le 17 \\ &x_1, x_2 \ge 0;\ \ x_1, x_2 \text{ integer} \end{aligned}$$

Problem 4 · LINGO: root LP (P1)

LINGO model

```
MAX = 4*x1 + 3*x2;
4*x1 + 9*x2 ≤ 26;
8*x1 + 5*x2 ≤ 17;
END
```

LINGO screenshot

save as `lingo/p4_P1.png`

Problem 5

Problem 5 · Statement

$$\begin{aligned} &\max\ z = 4x_1 + 5x_2 \\ \text{s.t.}\ \ &x_1 + 4x_2 \geq 5 \\ &3x_1 + 2x_2 \geq 7 \\ &x_1, x_2 \geq 0;\ \ x_1, x_2 \text{ integer} \end{aligned}$$

Problem 5 · LINGO: root LP (P1)

LINGO model

```
MAX = 4*x1 + 5*x2;
x1 + 4*x2 ≥ 5;
3*x1 + 2*x2 ≥ 7;
END
```

LINGO screenshot

save as `lingo/p5_P1.png`

Problem 6

Problem 6 · Statement

$$
\begin{aligned}
&\max\ z = 4x_1 + 5x_2 \\
\text{s.t.}\ \ &3x_1 + 2x_2 \le 10 \\
&x_1 + 4x_2 \le 11 \\
&3x_1 + 3x_2 \le 13 \\
&x_1, x_2 \ge 0;\ \ x_1, x_2 \text{ integer}
\end{aligned}
$$

Problem 6 · Subproblem P1: LINGO

LINGO model

```
MAX = 4*x1 + 5*x2;
3*x1 + 2*x2 <= 10;
x1 + 4*x2 <= 11;
3*x1 + 3*x2 <= 13;
END
```

Result

$z = \frac{187}{10} \approx 18.7, \quad x_1 = \frac{9}{5} \approx 1.8,$
$x_2 = \frac{23}{10} \approx 2.3$

Decision

$x_1 = \frac{9}{5}$ is fractional $\Rightarrow$ branch: **P2** with $x_1 \leq 1$ and **P3** with $x_1 \geq 2$.

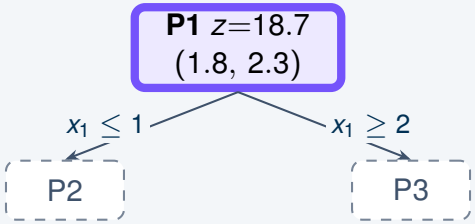
LINGO screenshot

save as `lingo/p6_P1.png`

Problem 6 · Tree after solving P1

Step 1: P1 solved (branched).

Current incumbent: none



Problem 6 · Subproblem P2: LINGO

LINGO model

```
MAX = 4*x1 + 5*x2;
3*x1 + 2*x2 <= 10;
x1 + 4*x2 <= 11;
3*x1 + 3*x2 <= 13;
x1 <= 1;
END
```

Result

$z = \frac{33}{2} \approx 16.5, \quad x_1 = 1, \quad x_2 = \frac{5}{2} \approx 2.5$

Decision

$x_2 = \frac{5}{2}$ is fractional $\Rightarrow$ branch: **P4** with $x_2 \leq 2$ and **P5** with $x_2 \geq 3$.

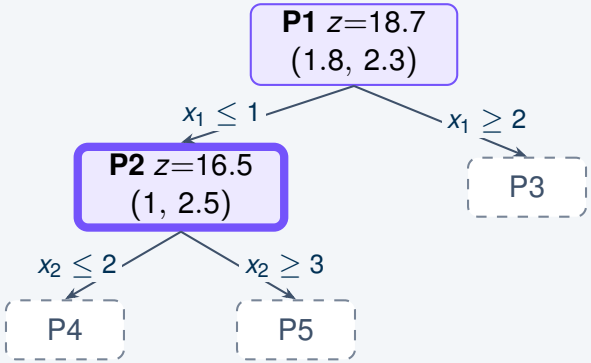
LINGO screenshot

save as `lingo/p6_P2.png`

Problem 6 · Tree after solving P2

Step 2: P2 solved (branched).

Current incumbent: none



Problem 6 · Subproblem P4: LINGO

LINGO model

```
MAX = 4*x1 + 5*x2;
3*x1 + 2*x2 ≤ 10;
x1 + 4*x2 ≤ 11;
3*x1 + 3*x2 ≤ 13;
x1 ≤ 1;
x2 ≤ 2;
END
```

Result

$z = 14, \quad x_1 = 1, \quad x_2 = 2$

Decision

Feasible integer solution $\Rightarrow$ new **incumbent** $z^* = 14$. Node fathomed.

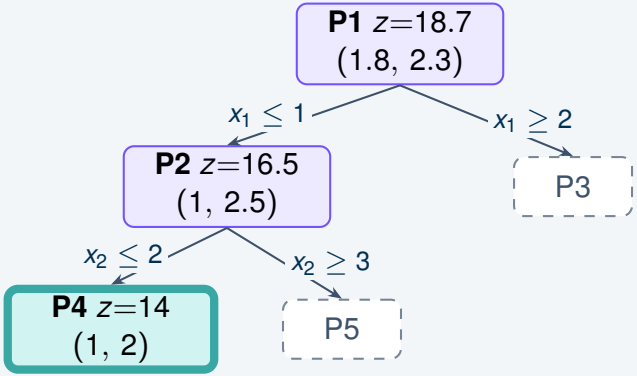
LINGO screenshot

save as `lingo/p6_P4.png`

Problem 6 · Tree after solving P4

Step 3: P4 solved (integer).

Current incumbent: $z^* = 14$ at $(1, 2)$



Problem 6 · Subproblem P5: LINGO

LINGO model

```
MAX = 4*x1 + 5*x2;
3*x1 + 2*x2 ≤ 10;
x1 + 4*x2 ≤ 11;
3*x1 + 3*x2 ≤ 13;
x1 ≤ 1;
x2 ≥ 3;
END
```

Result

LINGO reports: **Infeasible**

Decision

Empty feasible region ⇒ node fathomed by **infeasibility**.

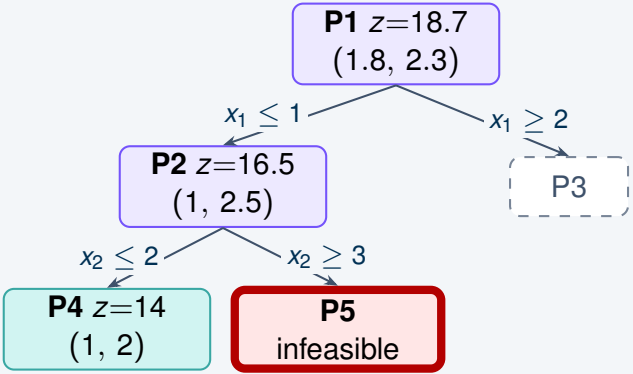
LINGO screenshot

save as `lingo/p6_P5.png`

Problem 6 · Tree after solving P5

Step 4: P5 solved (infeasible).

Current incumbent: $z^* = 14$ at $(1, 2)$



Problem 6 · Subproblem P3: LINGO

LINGO model

```
MAX = 4*x1 + 5*x2;
3*x1 + 2*x2 <= 10;
x1 + 4*x2 <= 11;
3*x1 + 3*x2 <= 13;
x1 >= 2;
END
```

Result

$z = 18, \quad x_1 = 2, \quad x_2 = 2$

Decision

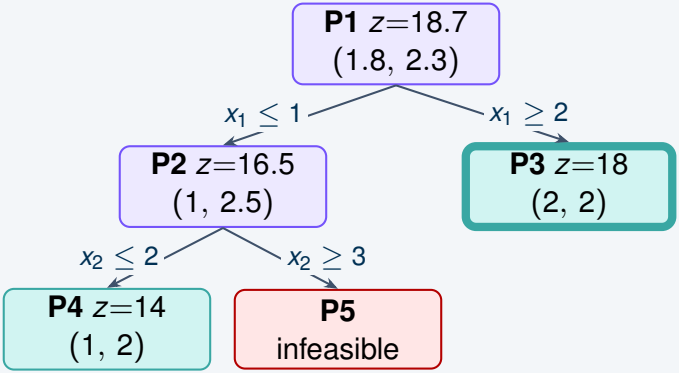
Feasible integer solution $\Rightarrow$ new **incumbent** $z^* = 18$. Node fathomed.

LINGO screenshot

save as `lingo/p6_P3.png`

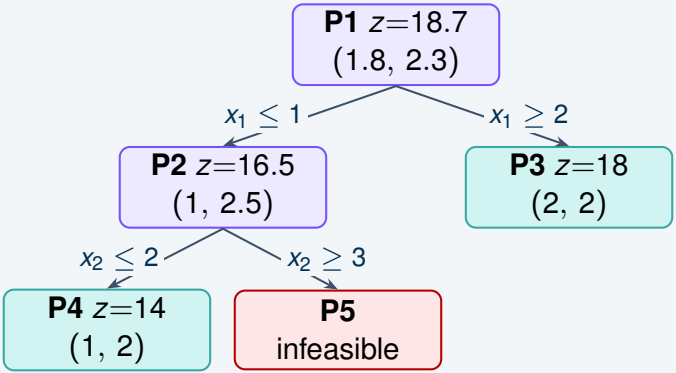
Problem 6 · Tree after solving P3

Step 5: P3 solved (integer). Current incumbent: $z^* = 18$ at $(2, 2)$



Problem 6 · Final solution

Optimal solution: $z^* = 18$, $x_1^* = 2$, $x_2^* = 2$ (5 subproblems solved)



Problem 7

Problem 7 · Statement

$$\begin{aligned} &\max\ z = 7x_1 + 3x_2 \\ \text{s.t.}\ \ &2x_1 + x_2 \le 9 \\ &3x_1 + 2x_2 \le 13 \\ &x_1, x_2 \ge 0;\ \ x_1, x_2 \text{ integer} \end{aligned}$$

Problem 7 · Subproblem P1: LINGO

LINGO model

```
MAX = 7*x1 + 3*x2;
2*x1 + x2 <= 9;
3*x1 + 2*x2 <= 13;
END
```

Result

$z = \frac{91}{3} \approx 30.33, \quad x_1 = \frac{13}{3} \approx 4.33, \quad x_2 = 0$

Decision

$x_1 = \frac{13}{3}$ is fractional $\Rightarrow$ branch: **P2** with $x_1 \leq 4$ and **P3** with $x_1 \geq 5$.

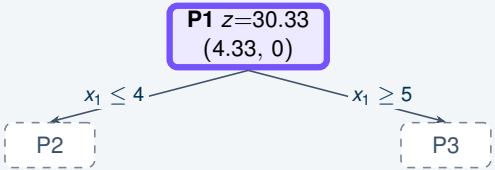
LINGO screenshot

save as `lingo/p7_P1.png`

Problem 7 · Tree after solving P1

Step 1: P1 solved (branched).

Current incumbent: none



Problem 7 · Subproblem P2: LINGO

LINGO model

```
MAX = 7*x1 + 3*x2;
2*x1 + x2 ≤ 9;
3*x1 + 2*x2 ≤ 13;
x1 ≤ 4;
END
```

Result

$z = \frac{59}{2} \approx 29.5, \quad x_1 = 4, \quad x_2 = \frac{1}{2} \approx 0.5$

Decision

$x_2 = \frac{1}{2}$ is fractional $\Rightarrow$ branch: **P4** with $x_2 \leq 0$ and **P5** with $x_2 \geq 1$.

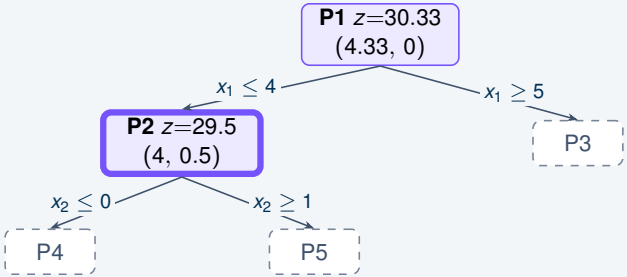
LINGO screenshot

save as `lingo/p7_P2.png`

Problem 7 · Tree after solving P2

Step 2: P2 solved (branched).

Current incumbent: none



Problem 7 · Subproblem P4: LINGO

LINGO model

```
MAX = 7*x1 + 3*x2;
2*x1 + x2 <= 9;
3*x1 + 2*x2 <= 13;
x1 <= 4;
x2 <= 0;
END
```

Result

$z = 28, \quad x_1 = 4, \quad x_2 = 0$

Decision

Feasible integer solution $\Rightarrow$ new **incumbent** $z^* = 28$. Node fathomed.

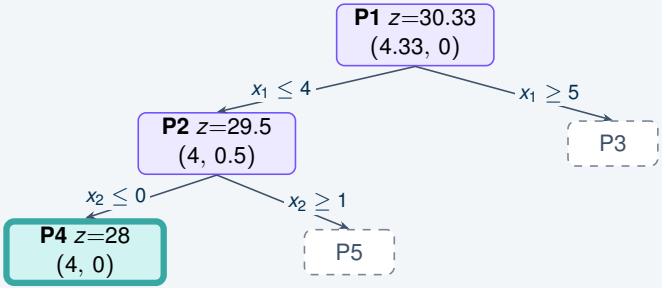
LINGO screenshot

save as `lingo/p7_P4.png`

Problem 7 · Tree after solving P4

Step 3: P4 solved (integer).

Current incumbent: $z^* = 28$ at $(4, 0)$



Problem 7 · Subproblem P5: LINGO

LINGO model

```
MAX = 7*x1 + 3*x2;
2*x1 + x2 ≤ 9;
3*x1 + 2*x2 ≤ 13;
x1 ≤ 4;
x2 ≥ 1;
END
```

Result

$z = \frac{86}{3} \approx 28.67, \quad x_1 = \frac{11}{3} \approx 3.67, \quad x_2 = 1$

Decision

$x_1 = \frac{11}{3}$ is fractional $\Rightarrow$ branch: **P6** with $x_1 \leq 3$ and **P7** with $x_1 \geq 4$.

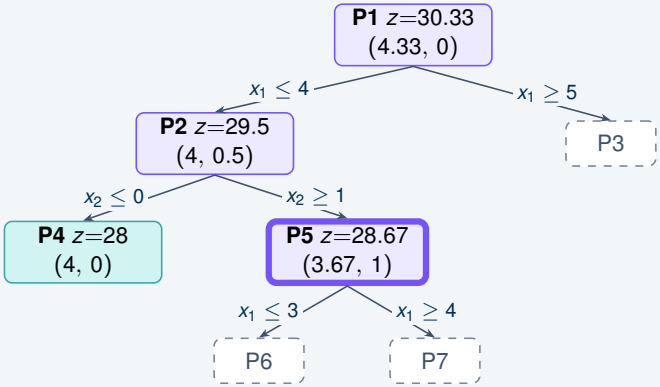
LINGO screenshot

save as `lingo/p7_P5.png`

Problem 7 · Tree after solving P5

Step 4: P5 solved (branched).

Current incumbent: $z^* = 28$ at $(4, 0)$



Problem 7 · Subproblem P6: LINGO

LINGO model

```
MAX = 7*x1 + 3*x2;
2*x1 + x2 ≤ 9;
3*x1 + 2*x2 ≤ 13;
x1 ≤ 3;
x2 ≥ 1;
END
```

Result

$z = 27, \quad x_1 = 3, \quad x_2 = 2$

Decision

$z = 27$ does not improve the incumbent $z^* = 28$ ($\leq$) $\Rightarrow$ pruned by **bound** (the solution is integer but not better).

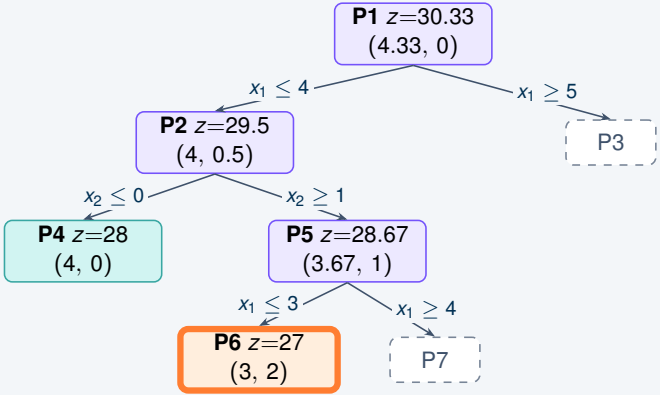
LINGO screenshot

save as `lingo/p7_P6.png`

Problem 7 · Tree after solving P6

Step 5: P6 solved (pruned).

Current incumbent: $z^* = 28$ at $(4, 0)$



Problem 7 · Subproblem P7: LINGO

LINGO model

```
MAX = 7*x1 + 3*x2;
2*x1 + x2 ≤ 9;
3*x1 + 2*x2 ≤ 13;
x1 ≤ 4;
x2 ≥ 1;
x1 ≥ 4;
END
```

Result

LINGO reports: **Infeasible**

Decision

Empty feasible region ⇒ node fathomed by **infeasibility**.

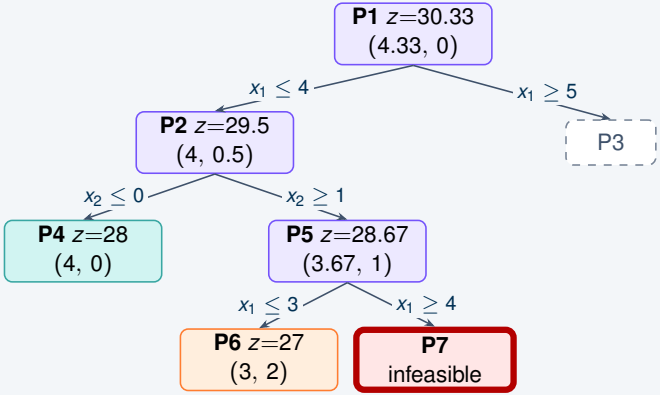
LINGO screenshot

save as `lingo/p7_P7.png`

Problem 7 · Tree after solving P7

Step 6: P7 solved (infeasible).

Current incumbent: $z^* = 28$ at $(4, 0)$



Problem 7 · Subproblem P3: LINGO

LINGO model

```
MAX = 7*x1 + 3*x2;
2*x1 + x2 ≤ 9;
3*x1 + 2*x2 ≤ 13;
x1 ≥ 5;
END
```

Result

LINGO reports: **Infeasible**

Decision

Empty feasible region ⇒ node fathomed by **infeasibility**.

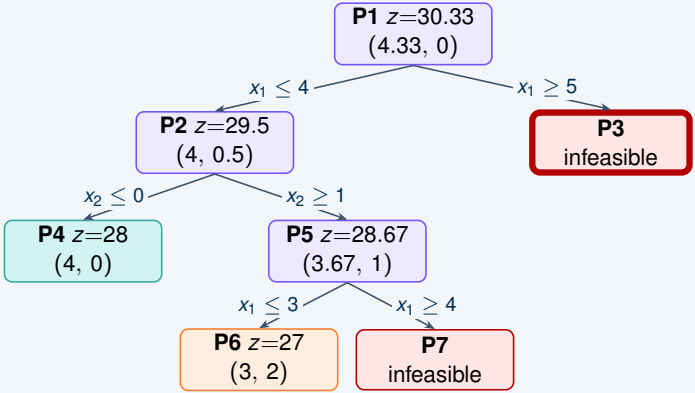
LINGO screenshot

save as `lingo/p7_P3.png`

Problem 7 · Tree after solving P3

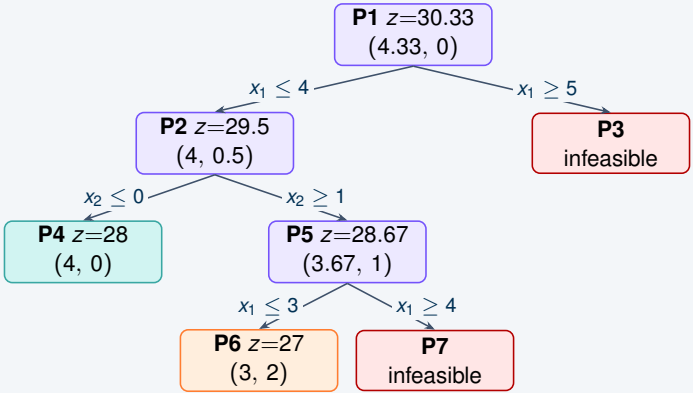
Step 7: P3 solved (infeasible).

Current incumbent: $z^* = 28$ at $(4, 0)$



Problem 7 · Final solution

Optimal solution: $z^* = 28$, $x_1^* = 4$, $x_2^* = 0$ (7 subproblems solved)



Summary

Summary of results

Problem	Objective	Subproblems	z^*	(x_1^*, x_2^*)
Problem 6	max	5	18	(2, 2)
Problem 7	max	7	28	(4, 0)

Investigación de Operaciones

Dayron Josué Porras Pérez (2023124693)
Javier Lee Liang (2020184824)
José Julián Araya Castillo (2020098909)

Integer Programming

November 8, 2026